



IIT ROORKEE



NPTEL ONLINE
CERTIFICATION COURSE

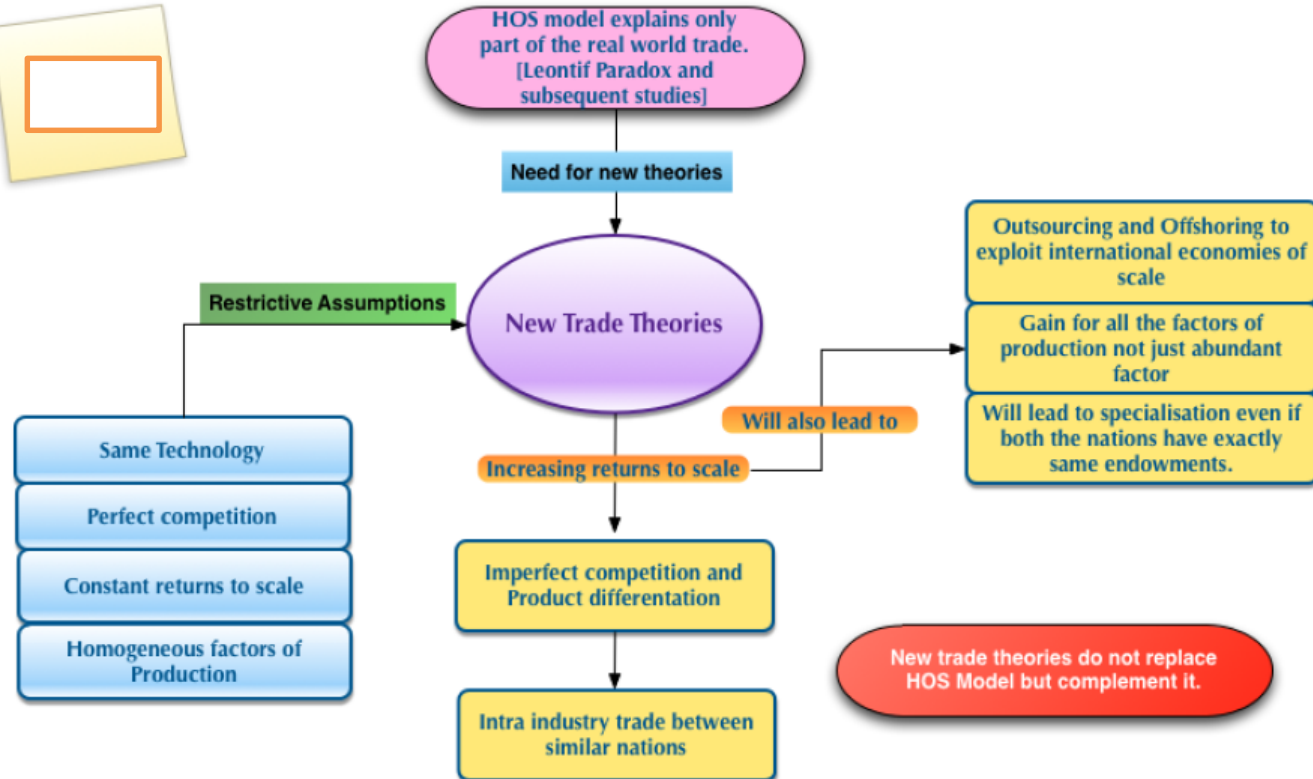
Imperfect Competition and Trade

Dr. PRATAP C. MOHANTY

DEPT. OF HUMANITIES & SOCIAL SCIENCES, IIT ROORKEE







Background

- Neoclassical model assumes industries are perfectly competitive, exhibit constant returns to scale and sell homogeneous goods.
- Neoclassical theory predicts trade will be interindustry in nature,
- however, there is empirical evidence that the structure of trade in manufactured goods is in part of an intra-industry nature.
- Analysis has shown that this type of trade has something to do with imperfectly competitive market structures, scale economies and differentiated goods.
- When defining comparative advantage, the Ricardian model and the Heckscher-Ohlin model both assume **constant returns to scale**:
 - If all factors of production are doubled then output will also double.

- But a firm or industry may have **increasing returns to scale** or **economies of scale**:
 - If all factors of production are doubled, then output will more than double.
 - Larger is more efficient: the cost per unit of output falls as a firm or industry increases output.
- The Ricardian and Heckscher-Ohlin models also rely on competition to predict that all income from production is paid to owners of factors of production: no “excess” or monopoly profits exist.
- But when economies of scale exist, large firms may be more efficient than small firms, and the industry may consist of a monopoly or a few large firms.
 - Production may be imperfectly competitive in the sense that excess or monopoly profits are captured by large firms.

Types of Economies of Scale

- Economies of scale could mean either that larger firms or that a larger industry (e.g., one made of more firms) is more efficient.
- **External economies of scale** occur when cost per unit of output depends on the *size of the industry*.
- **Internal economies of scale** occur when the cost per unit of output depends on the *size of a firm*.
- External economies of scale may result if a larger industry allows for more efficient provision of services or equipment to firms in the industry.
 - Many small firms that are competitive may comprise a large industry and benefit from services or equipment efficiently provided to the large group of firms.
- Internal economies of scale result when large firms have a cost advantage over small firms, which leads to an imperfectly competitive market.

Competition and Monopoly

- - Monopoly in one or more of the industries in the basic model results in a distortion,
- Abstracting from the reason for monopoly, and given constant returns to scale, assume the home country has a monopoly producer of good X.
- Under perfect competition: $p=mc$
- Hence, with competition in X and Y:
 - $p_x / p_y = MC_x / MC_y = MRT$
- For a monopolist: $TR = p_x X$
- Hence, change in revenue is: $dTR = p_x dX + X dp_x$

- Dividing by dX gives marginal revenue:
 - $MR_x = dTR/dX = p_x + (dp_x/dX).X$
 - shows that for a monopolist, marginal revenue will be less than price, given that $dp_x/dX < 0$
- Multiply second term of (5) by p_x / p_x , and factor out p_x :
 - $MR_x = p_x [1 + (dp_x/p_x) / (dX/X)]$
 - Term $(dp_x/p_x) / (dX/X) = 1/ex$, where ex is the price elasticity of demand for X
- Substituting in for the elasticity in: $MR_x = p_x [1 - 1/ex] = MC_x$

Figure 6-2

Average Versus Marginal Cost

This figure illustrates the average and marginal costs corresponding to the total cost function $C = 5 + x$. Marginal cost is always 1; average cost declines as output rises.

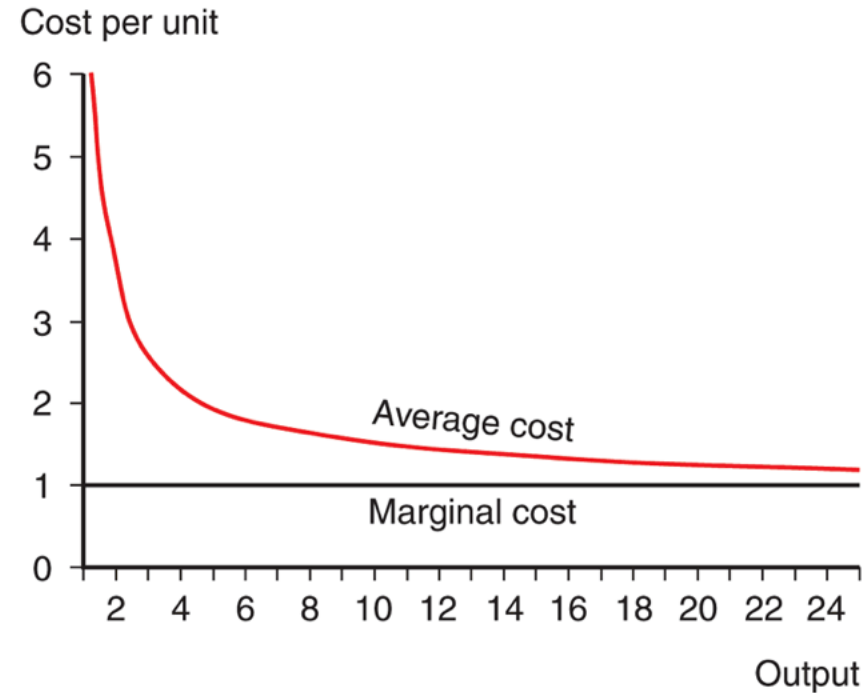
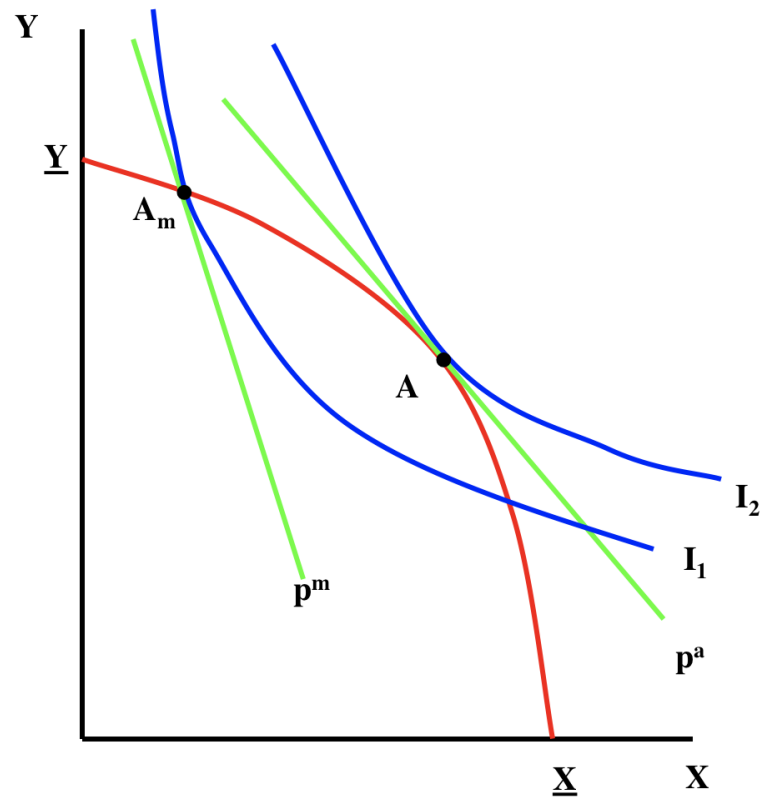
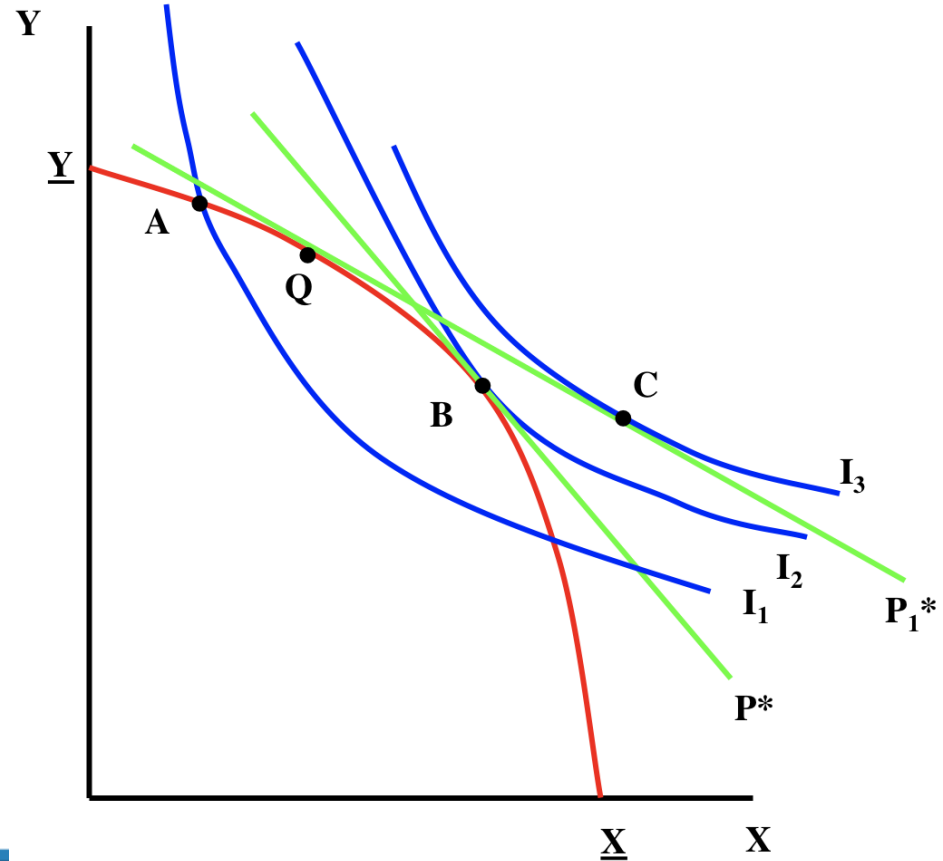


FIGURE 1: AUTARKY AND MONOPOLY



- i.e. under monopoly, $1/\epsilon_x$ measures the mark-up of price over marginal cost, so in general equilibrium:
 - $\{p_x [1 - 1/\epsilon_x]\} / p_y = MC_x / MC_y = MRT < p_x / p_y$
 - As $p_x > MC_x$, equilibrium price ratio p_x / p_y is greater than the slope of the production frontier
- A_m is the autarky equilibrium for the home country, given the autarky price ratio P_m :
 - output of X below competitive level at A
 - monopolist raises relative price of X above its competitive level at p_a
 - welfare is reduced below competitive level at A

FIGURE 2: PRO-COMPETITIVE GAINS FROM TRADE



- Distortion induced by monopoly is endogenous compared to say a production tax that raised X's price
- As the monopoly distortion is endogenous, trade may have additional benefits when there is imperfect competition - “pro-competitive” gains from trade
- autarky is at point A, X being monopolized; assuming this is a small country, it faces fixed world prices when it trades, which we assume are equal to undistorted autarky prices, $p_a = p^*$
- With trade, former monopolist faces a constant p_x^* , so $MR = p_x^*$, i.e. the perceived elasticity of demand is infinite, so monopoly distortion goes to zero

- Home country shifts to B, the move from A to B being the pure, pro-competitive gain from trade, i.e. the gain in a closed economy from eliminating monopoly
- Typically there will be gains due to comparative advantage, so world prices are p_1^* , and trade takes economy from A to C
- The gains are made up of the pro-competitive effect, A to B, and the normal gains from trade of B to C, i.e. the pro-competitive effect adds to the gains from trade

Monopolistic Competition

- **Monopolistic competition** is a model of an imperfectly competitive industry which assumes that
 1. Each firm can differentiate its product from the product of competitors.
 2. Each firm ignores the impact that changes in its own price will have on the prices competitors set: even though each firm faces competition it behaves as if it were a monopolist.

- A firm in a monopolistically competitive industry is expected:
 - to sell more the larger the total sales of the industry and the higher the prices charged by its rivals.
 - to sell less the larger the number of firms in the industry and the higher its own price.
- These concepts are represented by the mathematical relationship:

$$Q = S[1/n - b(P - P)]$$

- Q is an individual firm's sales
- S is the total sales of the industry
- n is the number of firms in the industry
- b is a constant term representing the responsiveness of a firm's sales to its price
- P is the price charged by the firm itself
- P is the average price charged by its competitors

- To make the model easier to understand, we assume that all firms have identical demand functions and cost functions.
 - Thus in equilibrium, all firms charge the same price: $P = P$
- In equilibrium,
 - $Q = S/n + 0$
 - $AC = C/Q = F/Q + c = F(n/S) + c$

1. *The Number of Firms and Average Cost*

$$AC = F(n/S) + c$$

- The larger the number of firms n in the industry, the higher the average cost for each firm because the less each firm produces.
- The larger the total sales S of the industry, the lower the average cost for each firm because the more that each firm produces.

2. *The Number of Firms and the Price*

$$Q = S[1/n - b(P - P)]$$

$$Q = S/n - Sb(P - P)$$

$$Q = S/n + SbP - SbP$$

$$Q = A - BxP$$

- Let $A \equiv S/n + SbP$ and $B \equiv Sb$

$$MR = P - Q/B = c$$

$$MR = P - Q/Sb = c$$

$$P = c + Q/Sb$$

$$P = c + (S/n)/Sb$$

$$P = c + 1/(nxb)$$

- The larger the number of firms n in the industry, the lower the price each firm charges because of increased competition.

- At some number of firms, the price that firms charge (which decreases in n) matches the average cost that firms pay (which increases in n).
- This number of firms is the number at which each firm has *zero profits*: price matches average cost.
- This number is the equilibrium number of firms.

- If the number of firms is greater than or less than n_2 , then the industry is not in equilibrium in the sense that firms have an incentive to exit or enter the industry.
 - Firms have an incentive to enter the industry when profits are greater than zero (price > average cost).
 - Firms have an incentive to exit the industry when profits are less than zero (price < average cost).

- Because trade increases market size, trade is predicted to decrease average cost in an industry described by monopolistic competition.
 - Industry sales increase with trade leading to decreased average costs:
 $AC = F(n/S) + c$
- Because trade increases the variety of goods that consumers can buy under monopolistic competition, it increases the welfare of consumers.
 - Because average costs decrease, consumers can also benefit from a decreased price.

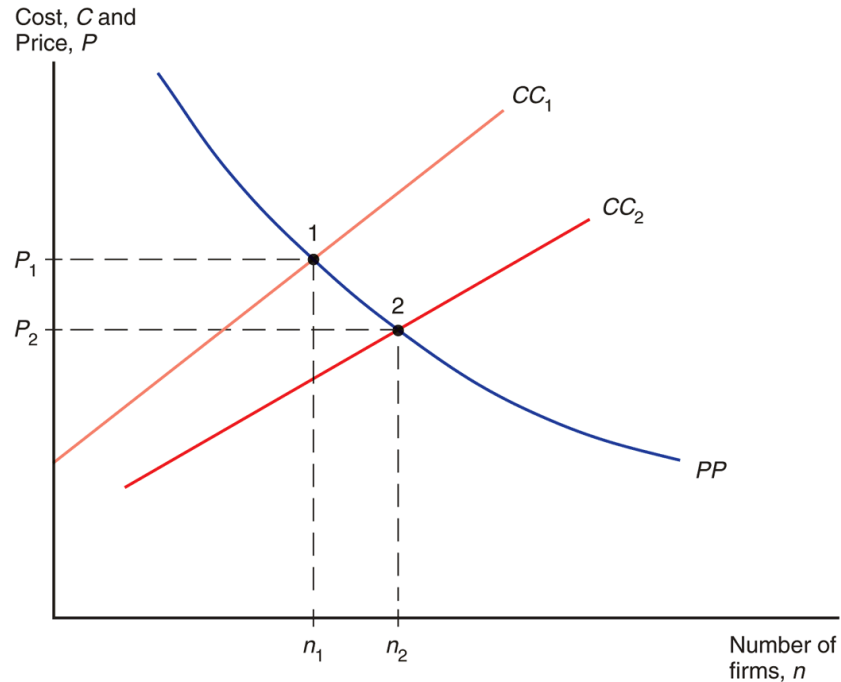


Figure 6-4

Effects of a Larger Market

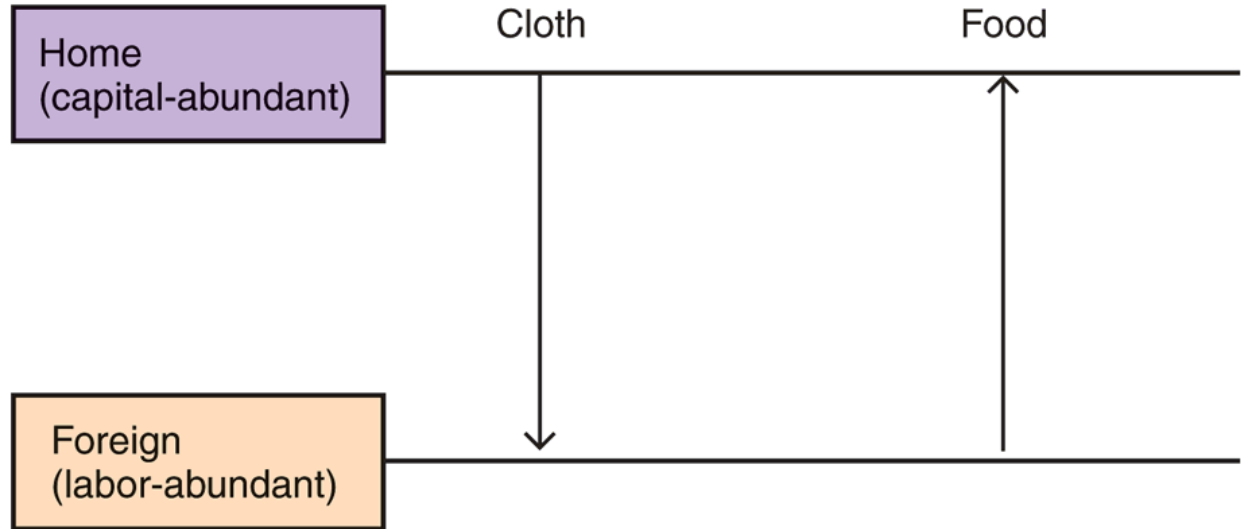
An increase in the size of the market allows each firm, other things equal, to produce more and thus have lower average cost. This is represented by a downward shift from CC_1 to CC_2 . The result is a simultaneous increase in the number of firms (and hence in the variety of goods available) and fall in the price of each.

Inter-industry Trade

Figure 6-6

**Trade in a World
Without Increasing
Returns**

In a world without economies of scale, there would be a simple exchange of cloth for food.



Intra-industry Trade

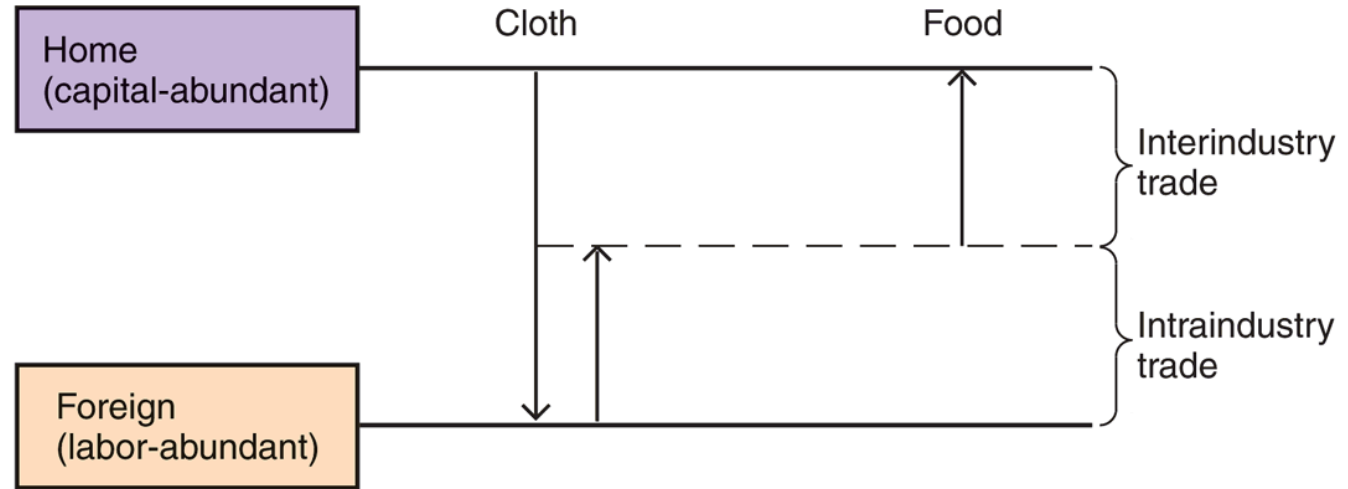


Figure 6-7

Trade with Increasing Returns and Monopolistic Competition

If cloth is a monopolistically competitive industry, Home and Foreign will produce differentiated products. As a result, even if Home is a net exporter of cloth goods, it will import as well as export cloth, giving rise to intraindustry trade.

*Thank
you*

